Majorana Modes in the Machine

Week 8: NISQ Reality Check – Noise Model and Failure Modes

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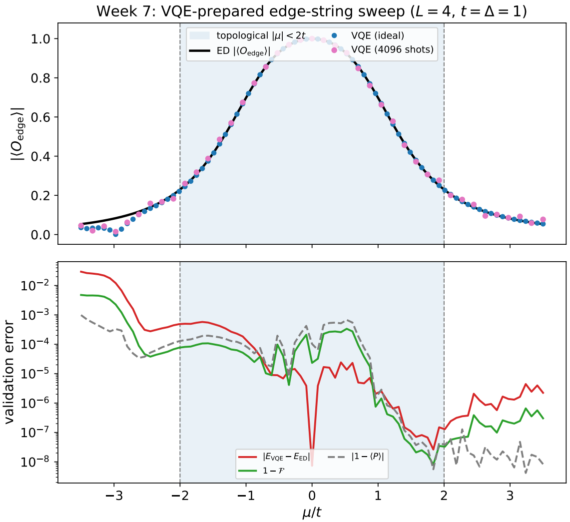
Week 8 Objective

Block 4 milestone

Start from the validated Week 7 VQE sweep and ask whether the edge-string phase diagnostic survives realistic circuit noise.

- Freeze the noiseless VQE result as the reference curve.
- Add noise to preparation and measurement circuits in controlled stages.
- Separate state-preparation failure from readout/estimation failure.
- Define when the inferred phase boundary becomes unreliable.

What Week 7 Gives Us



Noiseless baseline

- Prepared by parity-constrained VQE
- Edge string measured ideally and with shots
- ED validates energy, parity, fidelity, string value

This is the reference Block 4 must degrade in a physically interpretable way.

What Noise Can Break

State preparation

- VQE ansatz gates no longer implement exactly $U(\theta)$.
- Entangling-gate errors reduce edge coherence.
- Relaxation can bias the parity sector.

Measurement

- Readout flips corrupt the bitstring parity product.
- Shot noise broadens the string estimator.
- A small true signal becomes hard to distinguish from zero.

Main risk

Noise can flatten the edge-string contrast between $|\mu| < 2t$ and $|\mu| > 2t$, even when the noiseless circuit is correct.

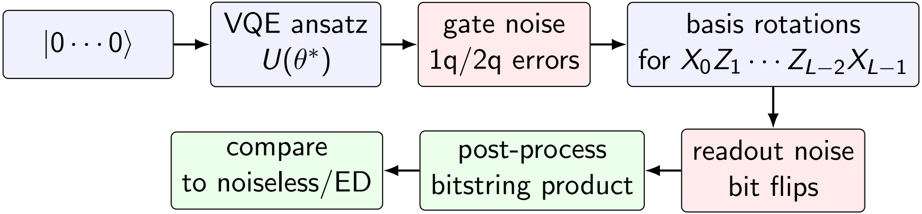
Noise Channels to Add

Channel	Physical meaning	Expected effect
Depolarizing error	Random Pauli error after gates	Shrinks correlations and string contrast
Readout error	Classical bit flip during measurement	Biases ± 1 parity-product estimator
Amplitude damping	T_1 relaxation toward $\lvert 0 \rangle$	Can change particle number and parity balance
Phase damping	T_2 dephasing	Destroys coherent edge-to-edge correlations

Week 8 scope

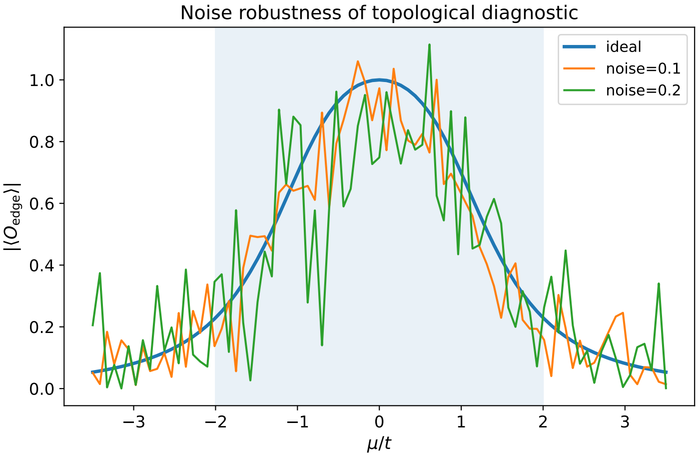
Build the protocol and first controlled tests; do not claim hardware realism until the noise parameters are tied to an explicit backend model.

Where Noise Enters the Circuit



The same observable is kept fixed; only the channel between preparation and bitstrings is changed.

First Sensitivity Sketch



This plot is a toy robustness check, not yet a hardware noise model.

- It perturbs the ideal edge-string curve directly.
- It shows the kind of contrast loss we must quantify.
- Week 8 replaces this with circuit-level noise channels.

Two Experiments to Keep Separate

Frozen-parameter noise test

- Use $\theta^*(\mu)$ from noiseless Week 7.
- Add noise only during circuit execution.
- Measures readout and circuit degradation at fixed preparation.

Noisy optimization test

- Optimize the VQE cost under noisy estimates.
- Tests whether the optimizer can still find good parameters.
- Harder and more expensive; should come second.

Engineering rule

First isolate measurement/circuit degradation; only then study optimizer degradation.

Week 8 Implementation Plan

```python
for mu, theta_star in noiseless_vqe_sweep:
    qc = build_preparation_circuit(theta_star)
    qc = append_edge_string_measurement(qc)

    for noise_level in noise_grid:
        backend = AerSimulator(noise_model=model(noise_level))
        counts = run(qc, backend, shots=shots)
        string = bitstring_product_average(counts)
        save(mu, noise_level, string)
```

- Reuse the Week 7 optimized parameters as the baseline input.
- Sweep one noise knob at a time before combining channels.
- Save raw counts, not only processed expectation values.

Metrics for Block 4

Metric	**Definition**	**Question**
String attenuation	$\lvert O_{\mathrm{noisy}}\rvert / \lvert O_{\mathrm{ideal}}\rvert$	How fast is the signal lost?
Phase contrast	topological mean $-$ trivial mean	Is the phase distinction still visible?
Boundary error	shift in inferred $\lvert\mu_c\rvert$	Does noise move the transition estimate?
Shot uncertainty	bootstrap/error bars from counts	Is the conclusion statistically resolved?
Parity drift	$\lvert 1 - \langle P \rangle \rvert$ if measurable	Did noise leave the target sector?

Definition of Done for Week 8

- A documented circuit-level noise protocol for the Week 7 VQE sweep.
- First frozen-parameter noisy edge-string curves for at least one noise channel.
- Comparison against the noiseless VQE and ED baselines.
- Clear separation between shot noise, readout noise, and gate noise.
- A quantitative criterion for when the phase-boundary diagnostic fails.

Next milestone

After the frozen-parameter tests, run noisy VQE optimization and compare raw versus mitigated readout.